PREPRINT · NOT PEER REVIEWED

Knots: seven mechanisms, what each would do, and a recording that would tell them apart

Eugene Jhong · Knots of Existence, independent · 28 September 2026 · draft 1 · field guide run 6231bcb2ebe1

Correspondence and objections: github.com/ejhong/knots/issues · PDF · the pilot protocol

ABSTRACT

People who work with the body report knots: small places that hold for months or years and let go within seconds, under a hand, with a slow breath or to attention, often with a tingle over the skin. No one has recorded one letting go. We set out seven mechanisms that could explain them: a small artery held shut where it pierces the fascia (perforators), vascular smooth muscle latched by a held prediction (the vascular latch), a myofascial trigger point, densified hyaluronan in the gliding layer between the fasciae, a sensitised nerve where it pierces the fascia, motor units latched on by persistent inward currents (a motor switch), and perception. Each was run as a model from its own physiology, with 60 parameters taken from a quoted line of a paper and 124 marked as guesses with ranges, through the same scenes (a stressful moment, slow breathing, attention, a resting hand, rolling, heat, an hour of moods) in 32 sampled settings each; what its knots did was read off the runs, not fitted to anyone's reports. The models part most at the moment of release: skin flow floods in (perforators), flow rises slowly (the latch), stiffness falls in the muscle (trigger points), the layers slide again (densification), a nerve's firing falls (nerves), a motor unit falls silent (the motor switch), or nothing changes in the tissue (perception). Set against practitioners' reports, the perforator model, extended with the sleeve of sliding tissue around each vessel, reproduces most of them, including release with a roller (18 of 19 settings), sparks (19 of 21), stiffness (12 of 19) and larger knots freeing smaller ones but not the reverse (5 of 5); releases to attention alone point instead to the latch (31 of 32) or the motor switch. Knots may be more than one thing. We propose a pilot recording at the moment of release, with laser speckle, single-motor-unit EMG and ultrasound against a pressed sham, whose outcome each model predicts in advance.

1. The problem

Massage therapists, physiotherapists and contemplative practitioners describe the same thing in different words: discrete, tender places in the soft tissue, most often over the neck, shoulders and back, that can persist for years and yet let go within seconds. Accounts from contemplative practice and bodywork add detail: knots seem to answer to the breath, letting go with a subtle, complex breath pattern; deeper ones to focused attention; a patient hand or a foam roller releases them one place at a time; a hot shower softens many at once; they gather where stress is held and limit movement; a release can come with a small pop or a tingle across a patch of skin, and another knot may move in nearby. These are impressions, not measurements, but they are specific enough to test.

The speed is the first clue. Collagen, scar and adhesion change over days and weeks, so whatever lets go within a breath must be a state the body is holding, not a structure it has built. At least seven candidates have been proposed or can be assembled from established physiology, and they disagree about where a knot is, what holds it and what releases it. None has been tested against a recording of a release, because none has been made.

Rather than choose among them, we ask each to show itself: to grow its own knots, from its own physiology, in the same piece of body under the same scenes, and to say how they would feel, respond and let go, and what an instrument would record. Where their knots differ in something a hand or an instrument can see, a measurement can decide.

2. Models and methods

2.1 One patch, the same scenes

Every model lives in the same 4 cm square of the upper back over the trapezius (skin, fat, the superficial and deep fascia, muscle), with stress held more toward the neck and at one place beside a spot where a hand rests and attention goes. Each places its own units where its anatomy puts them. The scenes are the same for all: rest, three minutes of stress, then a held level; thirty slow breaths (4 s in, 6 s out); the same with attention on the spot; subtle breaths with attention; a hand resting on the spot for a minute; a roller over a quiet place, and over the knot; five minutes of heat, as in a hot shower or a sauna; an hour of stress that wanders as moods do; half an hour more at the held level. Stress is one shared unit (0 rest, 1 a surge) mapped onto each theory's own inputs in a runner whose mapping is written out.

2.2 The seven models

Perforators. A small artery with tone in its wall is a switch: the pressure inside pushes it open, its muscle pulls it shut, and because muscle is strongest at one length there is a band of tone over which both states are stable [1, 2]. Parameters come from measured wall strength, thickness and pressure; the breath acts through the skin's sympathetic reflex [3] and through movement, which cuts smooth muscle's force [4]. Vessels form trees of a parent and its children sharing pressure, with conducted responses along their walls [5]. Each vessel crosses the gliding plane in a sleeve of sliding tissue [6] that its flow keeps wet; shut, the sleeve dries, its hyaluronan crowds (viscosity rising as the third or fourth power of concentration [7]), and it can jam. A knot is a vessel held shut by its own wall; a shut vessel starves its patch, and its nerve bursts as blood returns (the spark).

The vascular latch. Johnson's account [8]: a held prediction commands a region's small arteries; the contraction clamps the region and cuts it off from awareness, so the prediction cannot update. Formalised as a loop between the prediction and the clamp, set from the account's two claims (durable while unattended; not set by ordinary stress alone); attention or touch lets the prediction update. The latch-bridge state itself economises and does not remember [9].

Trigger points. The energy crisis [10]: endplate activity contracts sarcomeres, the contraction squeezes their capillaries, and starved sarcomeres cannot relax; sustained pressure lengthens the contracture over the tens of seconds of pressure release; a fast release twitches.

Densification. The gliding layer as a thixotropic material with a critical stress [11]: it rebuilds a structure at rest and loses it under shear, so between two drives both a jammed and a fluid state hold; a jammed patch gives way slowly and then all at once. Hyaluronan alone hardly changes with pH in the body's range [12] and does not gel [13], so a structure that jams must be more than hyaluronan.

Nerves. A nerve of the skin, sensitised where it pierces the fascia beside a perforator [14, 15], fires when pressed or stretched after a neuritis [16, 17] and when the sympathetic drive rises, about fourteen seconds late [18]. It has no switch: a knot is felt while its drive lasts.

The motor switch. Spinal motor neurons, once recruited, keep firing on a fraction of the input that recruited them, through persistent inward currents set by serotonin and noradrenaline [19, 20]; the metabolites of the contraction keep the loop going [21]. A knot is a latched unit; it unclenches at once when inhibited.

Perception. Central gain, arousal and attention turn ordinary input into a place felt as a knot, with a little hysteresis [22]. Nothing in the tissue changes.

2.3 What is felt and recorded

The same rules for all: a finger feels stiffness through skin and fat, weaker as $(a/\sqrt{(a^2 + d^2)})^3$ for a region of size a at depth d, above a threshold of touch; tenderness from each theory's own sensitiser; and what an instrument would record at the spot and at a pressed sham: skin flow by laser speckle, single motor units by surface EMG, stiffness by elastography, the layers' sliding by ultrasound, a nerve's firing.

2.4 Settings, scale and portraits

Each theory runs in 32 settings drawn by Sobol sampling across the ranges of its uncertain numbers. Every parameter carries its source and the quoted line, checked word for word against PubMed records, or is marked as a guess with a range. One shared rule sets each theory's scale (a stressful moment forms knots at the typical place; the held stress keeps them; the held stress alone does not form them); it is calibration, not a finding. Each theory's portrait is generated from the runs: a line holds in every setting (structural for that theory), in some (with the sampled number that best predicts where), or in none. The code and every number are open [23].

3. Results

3.1 What physics rules out

A liquid resists motion, not position: at its most viscous, hyaluronan around a vessel exerts a drag 12 times weaker than the pressure holding the vessel open [24, 25]. A patch of skin warms with a time constant of at least 44 s even at maximal skin flow, so warming cannot release a knot within a breath. Warmth thins hyaluronan by only about 2% a degree [7]. A vessel pressed shut cannot reopen until the pressure lifts. The sympathetic drive reaches a sensitised nerve only after about fourteen seconds [18]. And healthy small arteries hold themselves shut only over a band of tone (0.62–0.93 of maximal, rest 0.19); held shut at 4× resting tone, their muscle adapts to the new length and the knot sets, holding at rest, after about 1.7 h [26, 27].

3.2 The trace at release

The seven models part most sharply at the moment of release (Table 1). In every setting of each model the trace is its own; one recording at a release would separate them.

Table 1. What an instrument would record as a knot lets go, and what a hand would find, by model.

Model	At release	A finger finds	Under a resting hand	Attention alone
Perforators	skin flow floods in	nothing firm	as the hand lifts (never under it)	sometimes
Vascular latch	flow rises slowly	nothing firm	under the hand	often
Trigger points	stiffness falls in the muscle	a small nodule on a band, often too faint	under the hand	often
Densification	the layers slide again	nothing firm; the skin does not slide	not within a minute (never under it)	no
Nerves	the nerve's firing falls	nothing firm; a tender point	not within a minute	no
Motor switch	a motor unit falls silent	a firm band along the fibres	under the hand	often
Perception	nothing in the tissue	nothing firm	not within a minute (never under it)	sometimes

3.3 What a breath would have to do

However complex its pattern, a breath reaches a knot only through a few routes: the drive to its vessels or motor units, the tissue's movement, attention. For each model we found the least calming of the knot's own drive, held for a given time, that frees the typical knot (Figure 1). In no model does a single breath free the typical knot through drive; minutes of slow breathing do in some. The breath's movement, and attention where a model gives it a route, do what a single calming cannot.

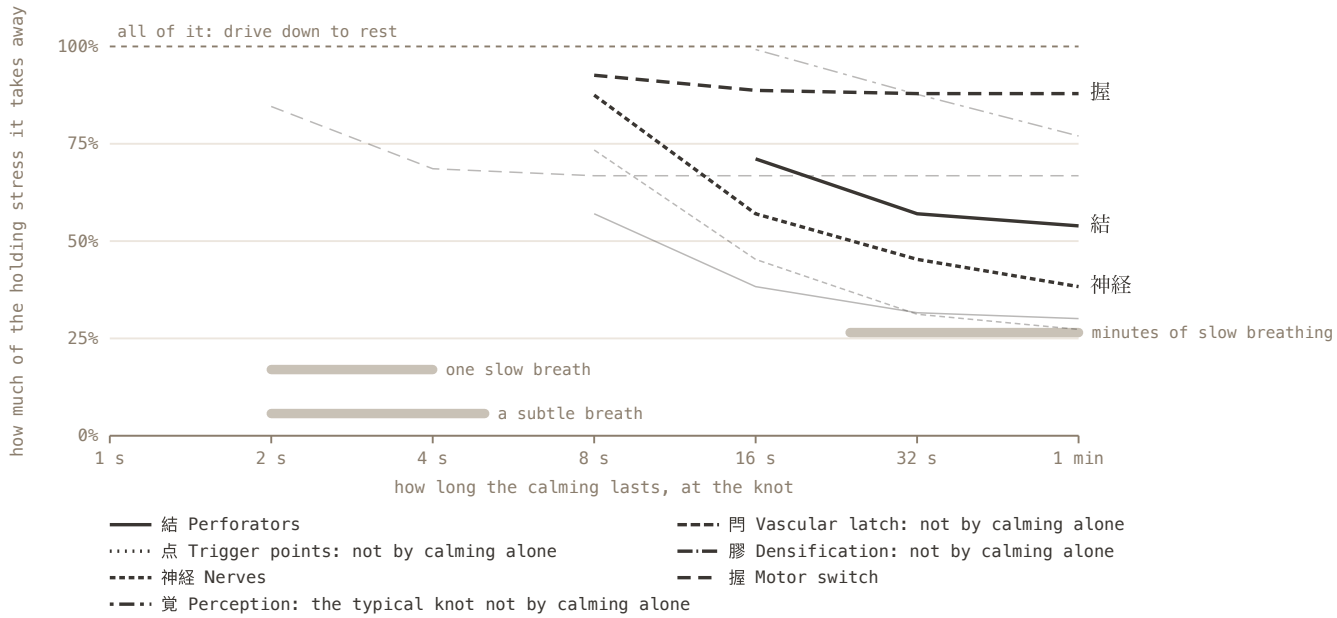


Figure 1. The least calming, as a share of the holding stress, that frees the typical knot (thick) and the easiest quarter (thin), against how long it lasts; grey bars: what one slow breath, a subtle breath and minutes of slow breathing do.

3.4 The perforator view in detail

Trees. Free one child under a held parent (its own drive down to rest for a minute): the child itself lets go in 5 of 18 settings, its parent in none of 18 settings, and a sibling in none of 18 settings. Free the parent instead: it lets go in 15 of 18 settings, and 2 of its 3 held children go with it (the median). Conducted dilation fades along the wall over about 2 mm [5], and a child's knot sits some 4-5 mm from its parent's.

Stiffness. Each vessel passes the fascia in a sleeve of sliding tissue that its flow keeps wet; shut, the sleeve dries, its hyaluronan crowds [7], and it can jam, pinning the layers there. Held half an hour more, the knot at the spot blocks a stretch in 12 of 19 settings; where it did not at first, the block comes after about 12 min. When such a knot lets go (its own drive calmed for a minute), its vessel opens after about 9 s; its sleeve slides again within ten minutes in none of 9 settings, and otherwise stays stuck until the layer is sheared (a roller, a moving hand).

Rolling and warmth. Rolled over, it lets go while the roller works, in 18 of 19 settings, after about 5 s. Each pass presses the vessel shut and lifts: it can reopen only in the lifts, while the squeezing weakens its wall's muscle [4] and the roller shears its sleeve. With five minutes of heat, as in a hot shower or a sauna, and the same slow breaths, the knot at the spot lets go where the breaths alone would not, or sooner, in 9 of 19 settings, after

about 80 s; across the patch about 100% let go (with the breaths alone, 74%). Heat lowers the tone of the vessels' walls [28], by the same number as it lowers the latch's clamp; what differs is the hold: once tone falls below its band, a shut vessel opens.

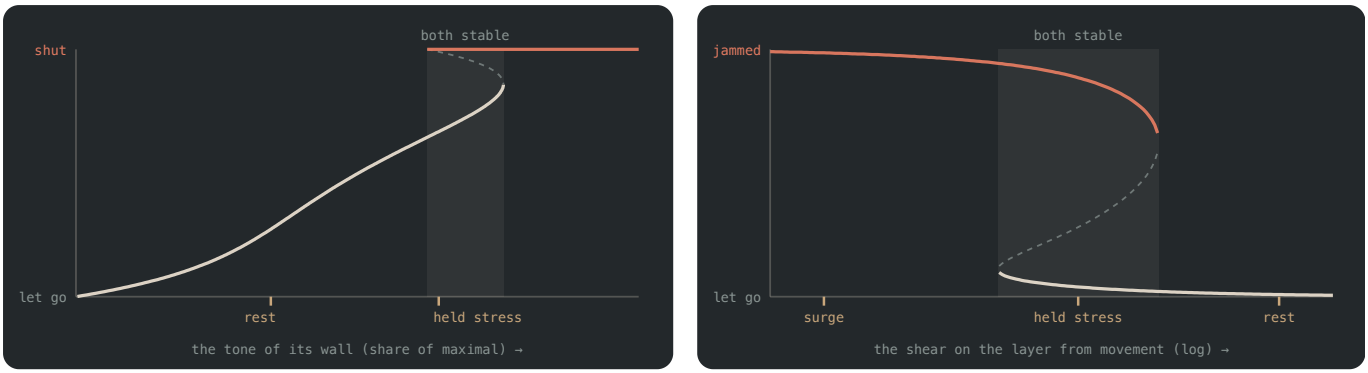


Figure 2. Two of the switches. Left: a perforator's wall, open or shut over a band of tone. Right: the gliding layer, fluid or jammed over a band of movement: a jammed patch stays jammed under movement that keeps a fluid one fluid.

3.5 The reports and the models

Only after the models had been run did we set practitioners' reports against them (Table 2). The perforator model, the most detailed of the seven, reproduces the most; attention alone points to the latch and the motor switch; the layers' stiffness is shared by densification, trigger points, the motor switch and the perforators' sleeves; four reports reach no model yet.

Table 2. In how many of the settings where it could happen each model's knots do what is reported (— no route).

Report	Perforators	Vascular latch	Trigger points	Densification	Nerves	Motor switch	Perception
They let go with the breath	16 of 19	2 of 32	12 of 22	12 of 32	20 of 32	17 of 32	14 of 32
Deeper ones let go to focused attention	9 of 19	31 of 32	12 of 22	0 of 32	0 of 32	16 of 32	15 of 32
A patient hand releases them, one at a time	8 of 19	18 of 32	8 of 22	0 of 32	0 of 32	20 of 32	7 of 32
A foam roller releases them	18 of 19	12 of 32	5 of 22	12 of 32	0 of 32	0 of 32	3 of 32
A hot shower softens many at once	6 of 19	12 of 32	–	0 of 32	–	–	–
They limit movement: a stretch meets a dull block	12 of 19	0 of 32	22 of 22	32 of 32	0 of 32	32 of 32	0 of 32
A pop, or a tingle across a patch of skin, as one goes	19 of 21	0 of 22	8 of 17	0 of 12	0 of 20	0 of 32	0 of 32
When one lets go, another moves in nearby	6 of 16	12 of 18	7 of 17	0 of 4	0 of 17	21 of 22	0 of 8
A big one frees small ones nearby, not the reverse	5 of 5	–	–	–	–	–	–
One comes back to roughly the same place	0 of 16	10 of 18	8 of 17	0 of 4	15 of 17	9 of 22	1 of 8
Working one side eases the other	–	–	–	–	–	–	–
Drinking water eases release	–	–	–	–	–	–	–
A release brings a wave of well-being	–	–	–	–	–	–	–
Layers peel apart, and fill in again	–	–	–	–	–	–	–

4. Discussion

More than one kind of knot. No single model reproduces every report. A perforator held shut, with its sleeve, accounts for the spark, the release with a roller or a lifting hand, stiffness that outlasts the release until the layer is moved, and larger knots freeing smaller ones; heat, once the breaths' own share is taken out, frees about as many latch knots as perforator knots. It does not account for knots that let go to attention alone, which a latched clamp or a latched motor unit would. The simplest reading is that practitioners meet at least two kinds of knot, and a recording should classify each release rather than crown one theory.

Perforators and latches: one muscle, two holds. Both are vascular smooth muscle held contracted, and heat reaches both through the same number in these models. They differ in what holds them: a perforator by its wall's mechanics (a band of tone over which open and shut are both

stable), the latch by information (a prediction that keeps its vessels clamped and the region out of awareness). They need not compete: a held prediction could be the command that keeps a perforator inside its band. Such a knot would let go to attention and to a roller alike, and spark as it went: the combination is the next model to build.

Peeling and filling. Some describe the fascia beneath the skin peeling apart along lines, leaving places that feel hollow, and then a filling-in felt as re-stitching. The perforator view suggests where to look: the perforators and their sleeves are the stitches that tether the superficial fascia to the deep; if many sleeves free at once and the returning flood fills the gliding plane, the layers could part along the lines between tethers, and close again as the fluid is taken up and the sleeves re-hold. This is not modelled; the fascia as a system of its own is the next model to build.

What would refute each. A structural line in a model, one that holds in every setting, is what a measurement can refute. A perforator knot that lets go while a hand still presses it; a latch knot that lets go with no rise in flow; a trigger point with no change in the muscle's stiffness; a densified patch whose layers slide as before; a nerve's knot a local anaesthetic cannot touch; a motor-switch knot with no unit falling silent; and, for perception, any change in the tissue that times itself to the felt release.

Limits. Most numbers specific to knots are guesses, marked and sampled; the models' traits say which of them decide each result. Three of the seven formalisations (the latch, densification, nerves) are ours, not yet checked by those who hold them. The shared scale gives every theory knots to study, by construction. The reports are rough impressions. A model shows what a mechanism would do with these numbers; it cannot show that knots are that mechanism.

Safety. Deep somatic and contemplative work can cause severe and lasting difficulties [29, 30]. Nothing here is a practice or a protocol for anyone to try; any recording belongs under ethics review, with ordinary breathing, comfortable pressure, and nothing on the front of the neck.

5. The measurement

A pilot of twelve participants with three to five knots each, about ninety minutes, gives forty to sixty release events, each marked by the participant with a button. At each: laser speckle over the knot and a pressed sham [31, 28]; high-density surface EMG over the upper trapezius, decomposed into single motor units [32, 33]; and ultrasound of the layers' sliding [34], before, at, and after the release and after the shoulder moves; with a blinded Doppler map of perforators under the knots. The predictions are fixed in advance by the models' runs. The protocol is on one page: the pilot.

Data, code and assistance

All models, parameters with their sources, runs and figures are open at https://github.com/ejhong/knots, and every figure and number on this page is generated from them (field guide run 6231bcb2ebe1). The models, the code and this text were developed with an AI assistant (Claude, Anthropic); the author is responsible for the content, and errors are welcome as issues.

References

1. Burton AC (1951). On the physical equilibrium of small blood vessels. *Am J Physiol 164:319-29*. doi.org/10.1152/ajplegacy.1951.164.2.319
2. Nichol J, Girling F, Jerrard W et al. (1951). Fundamental instability of the small blood vessels and critical closing pressures in vascular beds. *Am J Physiol 164:330-44*. doi.org/10.1152/ajplegacy.1951.164.2.330
3. Mayrovitz HN (2026). Physiological Impacts and Medical Applications of the Inspiratory Gasp Vascular Reflex. *Cureus 18:e105709*. doi.org/10.7759/cureus.105709
4. Ljung B, Sivertsson R (1975). Vibration-induced inhibition of vascular smooth muscle contraction. *Blood Vessels 12:38-52*. doi.org/10.1159/000158037
5. Segal SS, Damon DN, Duling BR (1989). Propagation of vasomotor responses coordinates arteriolar resistances. *Am J Physiol 256:H832-7*. doi.org/10.1152/ajpheart.1989.256.3.H832
6. Guimberteau JC, Sentucq-Rigall J, Panconi B et al. (2005). [Introduction to the knowledge of subcutaneous sliding system in humans]. *Ann Chir Plast Esthet 50:19-34*. doi.org/10.1016/j.anplas.2004.10.012
7. Cowman MK, Schmidt TA, Raghavan P et al. (2015). Viscoelastic Properties of Hyaluronan in Physiological Conditions. *F1000Res 4:622*. doi.org/10.12688/f1000research.6885.1
8. Michael Edward Johnson (2023). Principles of Vasocomputation: A Unification of Buddhist Phenomenology, Active Inference, and Physical Reflex (Part I). *opentheory.net, 12 July 2023 (updated 2025)*. opentheory.net/2023/07/principles-of-vasocomputation-a-unification-of-buddhist-phenomenology-active-inference-and-physical-reflex-part-i/
9. Rembold CM (1991). Relaxation, [Ca2+]i, and the latch-bridge hypothesis in swine arterial smooth muscle. *Am J Physiol 261:C41-50*. doi.org/10.1152/ajpcell.1991.261.1.C41
10. Gerwin RD, Dommerholt J, Shah JP (2004). An expansion of Simons' integrated hypothesis of trigger point formation. *Curr Pain Headache Rep 8:468-75*. doi.org/10.1007/s11916-004-0069-x
11. Coussot P, Nguyen QD, Huynh HT et al. (2002). Avalanche behavior in yield stress fluids. *Phys Rev Lett 88:175501*. doi.org/10.1103/PhysRevLett.88.175501
12. Gatej I, Popa M, Rinaudo M (2005). Role of the pH on hyaluronan behavior in aqueous solution. *Biomacromolecules 6:61-7*. doi.org/10.1021/bm040050m

13. Krause WE, Bellomo EG, Colby RH (2001). Rheology of sodium hyaluronate under physiological conditions. *Biomacromolecules 2:65-9*. doi.org/10.1021/bm0055798
14. Weum S, de Weerd L (2016). Perforator-Guided Drug Injection in the Treatment of Abdominal Wall Pain. *Pain Med 17:1229-32*. doi.org/10.1093/pm/pnv011
15. Quintner JL, Cohen ML (1994). Referred pain of peripheral nerve origin: an alternative to the "myofascial pain" construct. *Clin J Pain 10:243-51*. doi.org/10.1097/00002508-199409000-00012
16. Dilley A, Lynn B, Pang SJ (2005). Pressure and stretch mechanosensitivity of peripheral nerve fibres following local inflammation of the nerve trunk. *Pain 117:462-472*. doi.org/10.1016/j.pain.2005.08.018
17. Dilley A, Bove GM (2008). Resolution of inflammation-induced axonal mechanical sensitivity and conduction slowing in C-fiber nociceptors. *J Pain 9:185-92*. doi.org/10.1016/j.jpain.2007.10.012
18. Devor M, Jänig W, Michaelis M (1994). Modulation of activity in dorsal root ganglion neurons by sympathetic activation in nerve-injured rats. *J Neurophysiol 71:38-47*. doi.org/10.1152/jn.1994.71.1.38
19. Gorassini MA, Bennett DJ, Yang JF (1998). Self-sustained firing of human motor units. *Neurosci Lett 247:13-6*. doi.org/10.1016/s0304-3940(98)00277-8
20. Heckman CJ, Johnson M, Mottram C et al. (2008). Persistent inward currents in spinal motoneurons and their influence on human motoneuron firing patterns. *Neuroscientist 14:264-75*. doi.org/10.1177/1073858408314986
21. Johansson H, Sojka P (1991). Pathophysiological mechanisms involved in genesis and spread of muscular tension in occupational muscle pain and in chronic musculoskeletal pain syndromes: a hypothesis. *Med Hypotheses 35:196-203*. doi.org/10.1016/0306-9877(91)90233-o
22. Woolf CJ (2011). Central sensitization: implications for the diagnosis and treatment of pain. *Pain 152:S2-S15*. doi.org/10.1016/j.pain.2010.09.030
23. Knots of Existence: models, parameters and runs. https://github.com/ejhong/knots
24. Nicholls M, Manjoo A, Shaw P et al. (2018). A Comparison Between Rheological Properties of Intra-articular Hyaluronic Acid Preparations and Reported Human Synovial Fluid. *Adv Ther 35:523-530*. doi.org/10.1007/s12325-018-0688-y
25. Fede C, Angelini A, Stern R et al. (2018). Quantification of hyaluronan in human fasciae: variations with function and anatomical site. *J Anat 233:552-556*. doi.org/10.1111/joa.12866
26. Syyong H, Cheung C, Solomon D et al. (2008). Adaptive response of pulmonary arterial smooth muscle to length change. *J Appl Physiol (1985) 104:1014-20*. doi.org/10.1152/japplphysiol.01203.2007
27. Martinez-Lemus LA, Hill MA, Bolz SS et al. (2004). Acute mechanoadaptation of vascular smooth muscle cells in response to continuous arteriolar vasoconstriction: implications for functional remodeling. *FASEB J 18:708-10*. doi.org/10.1096/fj.03-0634fje
28. Minson CT, Berry LT, Joyner MJ (2001). Nitric oxide and neurally mediated regulation of skin blood flow during local heating. *J Appl Physiol (1985) 91:1619-26*. doi.org/10.1152/jappl.2001.91.4.1619
29. Lindahl JR, Fisher NE, Cooper DJ et al. (2017). The varieties of contemplative experience: A mixed-methods study of meditation-related challenges in Western Buddhists. *PLoS One 12:e0176239*. doi.org/10.1371/journal.pone.0176239
30. Farias M, Maraldi E, Wallenkampf KC et al. (2020). Adverse events in meditation practices and meditation-based therapies: a systematic review. *Acta Psychiatr Scand 142:374-393*. doi.org/10.1111/acps.13225
31. Roustit M, Millet C, Blaise S et al. (2010). Excellent reproducibility of laser speckle contrast imaging to assess skin microvascular reactivity. *Microvasc Res 80:505-11*. doi.org/10.1016/j.mvr.2010.05.012
32. Negro F, Muceli S, Castronovo AM et al. (2016). Multi-channel intramuscular and surface EMG decomposition by convolutive blind source separation. *J Neural Eng 13:026027*. doi.org/10.1088/1741-2560/13/2/026027
33. Dideriksen JL, Holobar A, Falla D (2016). Preferential distribution of nociceptive input to motoneurons with muscle units in the cranial portion of the upper trapezius muscle. *J Neurophysiol 116:611-8*. doi.org/10.1152/jn.01117.2015
34. Langevin HM, Fox JR, Koptiuch C et al. (2011). Reduced thoracolumbar fascia shear strain in human chronic low back pain. *BMC Musculoskelet Disord 12:203*. doi.org/10.1186/1471-2474-12-203